

BALRAM TIWARI

Germany · Available for relocation and remote/hybrid roles

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Summary

Machine Learning Engineer with experience building production-ready ML pipelines, model inference services, and NLP/LLM applications. Skilled in data preprocessing, feature engineering, model training, evaluation, deployment, and experiment tracking using Python, FastAPI, Docker, MLflow, and CI/CD workflows. Experienced in RAG pipelines, prompt engineering, knowledge graph construction, and ML applications in data-sensitive environments.

Technical Skills

Programming: Python, Java, SQL

Machine Learning: Scikit-learn, XGBoost, Classification, Feature Engineering, Model Evaluation, Baseline Evaluation, Hyperparameter Tuning

LLM & Generative AI: Retrieval-Augmented Generation (RAG), Prompt Engineering, Knowledge Graph Construction

Deep Learning & NLP: PyTorch, Transformers, HuggingFace, Named Entity Recognition (NER), Relation Extraction, BERT, LLaMA

ML Engineering & MLOps: FastAPI, Docker, MLflow, Model Deployment, REST APIs, CI/CD, Technical Documentation

Data & Operations: Pandas, NumPy, Data Preprocessing, ETL Pipelines, Monitoring

Cloud & Tools: AWS, Git, Linux, Streamlit, Jenkins

Languages: English (C1), German (A2)

Experience

Machine Learning Engineer (Working Student)

Jul 2024 – Present

DICE Group – University of Paderborn

Germany

- Built LLM/NLP pipelines for entity extraction, relation extraction, and structured knowledge graph generation from unstructured text.
- Implemented RAG-based extraction workflows using retrieval, context injection, prompt engineering, response parsing, and validation logic.
- Defined evaluation workflows for entity, relation, and subject-predicate-object triple extraction from Wikidata and DBpedia datasets.
- Fine-tuned transformer-based models using PyTorch and HuggingFace, improving NER F1-score from 0.31 to 0.65.

Machine Learning Engineer (Working Student)

Feb 2023 – Apr 2024

DEVK Versicherungen

Germany

- Developed a production-oriented ML pipeline for cross-sell prediction using customer, policy, demographic, and payment history data, covering preprocessing, feature engineering, model training, and inference.
- Trained and evaluated Logistic Regression and XGBoost models to estimate customer purchase probability for additional insurance products.
- Tracked experiments, model metrics, and validation results using MLflow, cross-validation, and classification metrics.
- Deployed the trained model as a FastAPI service to generate purchase probability scores for targeted marketing use cases.
- Documented pipeline assumptions, deployment steps, and data handling considerations for operational handover.

Java Software Engineer

Jul 2019 – Mar 2022

Capgemini

India

- Developed backend services and REST APIs using Java and Spring for enterprise financial services applications.
- Worked with Git, Maven, and Jenkins to support CI/CD workflows, automated builds, deployment pipelines, and release handover.
- Collaborated with DevOps teams to integrate applications into production deployment pipelines and maintain stable, monitored releases.

Projects

Time Series Forecasting using Machine Learning

- Developed time-series forecasting models using LSTM, XGBoost, and Prophet for financial time-series prediction.
- Performed feature engineering, baseline comparison, model evaluation, and hyperparameter tuning to improve forecasting performance.

Education

University of Koblenz

Master of Science – Web and Data Science

Apr 2022 – Present

Germany

Noida Institute of Engineering & Technology

Bachelor of Technology – Computer Science

Apr 2015 – Aug 2019

India

Publication

AAAI Conference: Contrastive Explanations for ABox Entailments

arXiv

- Proposed contrastive explanation methods for ontology-based reasoning systems to improve interpretability of AI decisions.